AN IMMERSION THAT IS QUASI-COMPACT OR HAS REDUCED SOURCE IS AN OPEN IMMERSION ONTO A SCHEME THEORETICALLY DENSE OPEN OF ITS SCHEME THEORETIC IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 1138-1146 and proof span 1148-1170 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof distinguishes the two hypotheses of the statement with the author's phrases 'In the case that Z is reduced' and 'In the case that Z to X is quasi-compact'; both are preserved.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The factorization of an immersion as an open immersion followed by a closed immersion, in the quasi-compact case and in the reduced case, cited from the same chapter. Cited at proof lines 3365-3367. Produces the closed intermediate scheme Z-bar-one inside which the source sits as an open subscheme.
- The scheme theoretic image of a morphism as the smallest closed subscheme through which it factors, and its density properties in the quasi-compact case, cited from the same chapter. Cited at proof lines 3368-3370 and 3384-3387. Places the scheme theoretic image inside the intermediate closed subscheme and supplies scheme theoretic and topological density.
- A closed immersion into a reduced scheme that is a bijection on underlying topological spaces is an isomorphism. Used at proof lines 3377-3381. Identifies the intermediate closed subscheme with the scheme theoretic image in the reduced case.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01QV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2280–2280. Same-chapter reference used in the retained author text at source line(s) 1149. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03DQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4429–4429. Same-chapter reference used in the retained author text at source line(s) 1150, 1159. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01R8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2293–2293. Same-chapter reference used in the retained author text at source line(s) 891, 903, 920, 928, 1007, 1084, 1167, 1169, 1369, 4982, 10102, 11100, 14144. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $h : Z \to X$ be an immersion. Assume either h is quasi-compact or Z is reduced. Let $\overline{Z} \subset X$ be the scheme theoretic image of h. Then the morphism $Z \to \overline{Z}$ is an open immersion which identifies Z with a scheme theoretically dense open subscheme of $\overline{Z}$. Moreover, Z is topologically dense in $\overline{Z}$.*

Proof. By Lemma 01QV or Lemma 03DQ we can factor $Z \to X$ as $Z \to \overline{Z}_1 \to X$ with $Z \to \overline{Z}_1$ open and $\overline{Z}_1 \to X$ closed. On the other hand, let $Z \to \overline{Z} \subset X$ be the scheme theoretic image of $Z \to X$. We conclude that $\overline{Z} \subset \overline{Z}_1$. Since Z is an open subscheme of $\overline{Z}_1$ it follows that Z is an open subscheme of $\overline{Z}$ as well. In the case that Z is reduced we know that $Z \subset \overline{Z}_1$ is topologically dense by the construction of $\overline{Z}_1$ in the proof of Lemma 03DQ. Hence $\overline{Z}_1$ and $\overline{Z}$ have the same underlying topological spaces. Thus $\overline{Z} \subset \overline{Z}_1$ is a closed immersion into a reduced scheme which induces a bijection on underlying topological spaces, and hence it is an isomorphism. In the case that $Z \to X$ is quasi-compact we argue as follows: The assertion that Z is scheme theoretically dense in $\overline{Z}$ follows from Lemma 01R8 part (3). The last assertion follows from Lemma 01R8 part (4). □